

Asymptotic $\mathfrak{su}(2)$ -Isotropy of the Effective Quadratic Form and the Identification $A_H = 2$

Q10 of the Cosmochrony Spectral Geometry Programme

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Abstract

Papers Q7–Q9 of the Cosmochrony spectral geometry programme establish that the effective operator L_{eff} on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ has a principal symbol $\sigma_2(L_{\text{eff}})|_{H_{\text{eff}}} = A_H(k_X^2 + k_Y^2) + A_Z k_Z^2$ with $A_H, A_Z > 0$ and $A_Z = 2 = C_{\mathfrak{su}(2)}$ (Casimir eigenvalue on $\text{Sym}^2(V_\rho)$, Q8). The identification $A_H = 2$ would complete the effective metric to $g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2)$ and establish the isotropic Lorentzian geometry of the Cosmochrony framework. The present paper proves $A_H \rightarrow 2$ from two structural inputs: (i) the asymptotic character-independence of the O-series spectral observables ($\sigma_c(n) \rightarrow \sigma_*(n)$ uniformly in c as $q \rightarrow \infty$, established structurally from BI parity [3, 6] and confirmed numerically by the O25 campaign [5]); and (ii) the uniqueness of the $\mathfrak{su}(2)$ -invariant quadratic form on $\text{Sym}^2(V_\rho)$ (Q7 Lemma 4.3 [9]). The core argument is that character-independence forces the effective form on H_{eff} to be scalar under the $\mathfrak{su}(2)$ action, and the unique such form is the Casimir with value 2. The result is established conditionally on the O-series universality (structurally supported, numerically confirmed for $q \leq 211$) and the bridge non-obstruction hypothesis of Q7–Q9.

Keywords. Heisenberg group; Weil representation; Born–Infeld bound; spectral universality; $\mathfrak{su}(2)$ -isotropy; Casimir operator; effective Lorentzian metric; Mosco convergence; Cosmochrony.

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1 Introduction

1.1 The remaining open problem

The chain of results Q5a–Q9 [2, 1, 9] reduces the geometry of the Cosmochrony effective spacetime to a single open question: does the horizontal coefficient A_H of the effective spatial co-metric equal the central coefficient $A_Z = 2$? More precisely, Q7–Q8 establish:

$$\sigma_2(L_{\text{eff}})|_{H_{\text{eff}}} = A_H(q)(k_X^2 + k_Y^2) + 2k_Z^2, \quad A_H(q) > 0, \quad A_Z = 2, \quad (1)$$

where $A_H(q)$ is determined numerically and $A_Z = C_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 2$ is fixed algebraically [4]. The bridge conjecture (Q7 Conjecture 4.7, [9]) is equivalent, given Q7–Q9, to the question:

$$A_H(q) \xrightarrow{q \rightarrow \infty} 2. \quad (2)$$

Numerical evidence from Q7 (Table 1) gives $|A_H(q) - 2| \approx 0.264 q^{-0.44}$ for $q \in \{61, 101, 151\}$, supporting (2); the present paper provides the structural argument for this convergence.

1.2 Main result

Theorem 1.1 (Asymptotic isotropy $A_H \rightarrow 2$). *Assume the O-series spectral universality [U] (Definition 3.1) as a convergence statement: $\varepsilon(q) \rightarrow 0$. Items (i)–(iii) of Proposition 3.2 provide structural support for [U]; item (iv) provides numerical evidence. Under [U] and the Q5a hypotheses [H1], [H2], [H-w], [H-E1], [C], the horizontal coefficient satisfies*

$$\lim_{q \rightarrow \infty} A_H(q) = 2. \quad (3)$$

More precisely, $|A_H(q) - 2| \leq C \varepsilon(q)$ for a constant $C > 0$ independent of q .

The value α is estimated numerically as ≈ 0.44 from O25 (Section 6). Together with Q7–Q9, Theorem 1.1 establishes the isotropic effective metric $g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2)$ and resolves Case 1 of the bridge conjecture.

1.3 Structure of the argument

The proof runs through three stages. First (Section 3), we establish that the O-series spectral observables $\sigma_c(n)$ become character-independent as $q \rightarrow \infty$: all characters $c \in (\mathbb{Z}/q\mathbb{Z})^\times$ produce the same fingerprint energy profile. Second (Section 4), we show that character-independence implies that the effective quadratic form on $H_{\text{eff}} = \text{Sym}^2(V_\rho)$ is asymptotically invariant under the $\mathfrak{su}(2)$ action. Third (Section 5), we use the uniqueness of the $\mathfrak{su}(2)$ -invariant quadratic form on the spin-1 representation (Casimir, value 2) to conclude $A_H \rightarrow 2$.

1.4 Organisation

Section 2 recalls the relevant results of Q7–Q9 and fixes notation. Section 3 states and establishes the O-series universality hypothesis [U]. Section 4 derives the asymptotic $\mathfrak{su}(2)$ -isotropy of the effective form. Section 5 completes the identification $A_H = 2$. Section 6 records the numerical confirmation from the O25 campaign. Section 7 draws the programme-level consequences.

2 Setup and Recall

2.1 The effective operator and its symbol

Let $L_{\text{eff}} = -A_\tau \partial_\tau^2 + A_H \Delta_H$ be the effective operator on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$, established conditionally in Q5b [1] and unconditionally (under Q5a hypotheses) by Q9 [8]. The effective subspace is $H_{\text{eff}} = \text{Sym}^2(V_\rho)$, where V_ρ is the spin- $\frac{1}{2}$ representation of $\mathfrak{su}(2)$ and $\text{Sym}^2(V_\rho)$ is the spin-1 representation with weights $m \in \{-1, 0, +1\}$ and spanning vectors $\{e_-, e_0, e_+\}$. By Q7 Proposition 6.1 [9] (no cross terms) and Q8 (Casimir rigidity, [4]), the restricted symbol takes the form (1) with $A_Z = 2$ and $A_H(q) > 0$ (Q9 Corollary 5.2, [8]).

2.2 O-series spectral observables

The O-series [3, 6, 7, 5] defines, for each prime q , character $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, and BFS depth n , the normalised Weil fingerprint energy:

$$\sigma_c(n) := \frac{\delta r_n}{|S_n|}, \quad (4)$$

where δr_n is the spectral radius increment and $|S_n|$ is the shell size at depth n . The pair observable is

$$\sigma_{\text{pair}}(n) := \sigma_c(n) \cdot \sigma_{q-c}(n), \quad (5)$$

with exponent δ_{pair} defined by $\sigma_{\text{pair}}(n) \sim n^{-\delta_{\text{pair}}}$ over the fitting window $[n_0, n_1]$. The BI-parity symmetry (O22 [3]):

$$\sigma_c(n) = \sigma_{q-c}(n) \quad \text{for all } c \text{ and all } n, \quad (6)$$

holds structurally and is confirmed to numerical precision across all tested primes.

2.3 The weight–character correspondence

The bridge identification of Q7 [9] maps the spin weights of $\text{Sym}^2(V_\rho)$ to O-series character sectors:

$$m = +1 \leftrightarrow c \in (0, q/2), \quad m = 0 \leftrightarrow c \in [q/2 - w, q/2 + w], \quad m = -1 \leftrightarrow c \in (q/2, q). \quad (7)$$

This correspondence is the content of Q7 Remark 4.2 [9]: the three spin sectors of $\text{Sym}^2(V_\rho)$ (the horizontal $e_\pm$ and central e_0 directions) are identified with the character sectors of the Weil fingerprint through the bridge φ . The central sector $m = 0$ is identified with a symmetric window of characters of the form $c \in [q/2 - w, q/2 + w]$ for a fixed width $w \geq 1$ independent of q ; its contribution under the uniform bound of [U] is therefore comparable to that of any other character, since $|\sigma_c(n) - \sigma_*(n)| \leq \varepsilon(q)\sigma_*(n)$ holds uniformly in c . Under this identification, the coefficient $A_H(q)$ measures how the fingerprint energy varies across the $m = \pm 1$ sectors relative to the $m = 0$ sector ($A_Z = 2$).

3 O-Series Universality

3.1 The universality hypothesis

Definition 3.1 (O-series spectral universality, [U]). The O-series satisfies spectral universality [U] if the fingerprint energy $\sigma_c(n)$ becomes character-independent as $q \rightarrow \infty$:

there exists a function $\sigma_*(n)$ such that

$$\max_{c \in (\mathbb{Z}/q\mathbb{Z})^\times} |\sigma_c(n) - \sigma_*(n)| \leq \varepsilon(q) \sigma_*(n), \quad (8)$$

where $\varepsilon(q) \rightarrow 0$ as $q \rightarrow \infty$, uniformly over $n \leq n^*(q)$.

The function $\sigma_*(n)$ is the universal power law $\sigma_*(n) = C_* n^{-\delta_{\text{pair}}/2}$ with $\delta_{\text{pair}} \approx 7.44$ (O-series canonical value).

3.2 Structural support for [U]

Proposition 3.2 (Structural support for [U]). *The O-series universality hypothesis [U] holds in the following structural sense.*

- (i) (BI-parity, O22 [3]). *The symmetry $\sigma_c(n) = \sigma_{q-c}(n)$ holds unconditionally. Consequently, the $m = +1$ and $m = -1$ sectors are exactly equal: $\sigma_c = \sigma_{q-c}$ for all c and q .*
- (ii) (Quaternionic minimality, O23 [6]). *The pair structure $\Sigma_c(n_3) = 3$ (three stable directions) is derived from the BI-parity constraint and holds unconditionally. This fixes the structure of $\sigma_*(n)$ within the fitting window.*
- (iii) (Rank stability, O24 [7]). *The observable rank is stable across conjugate pairs $(c, q - c)$ with the same exponent $\delta_{\text{pair}}(c) = \delta_{\text{pair}}(q - c)$ for all tested primes $q \in \{29, 61, 101, 151\}$.*
- (iv) (Numerical universality, O25 [5]). *The per-pair exponents satisfy $\text{Var}_c[\delta_{\text{pair}}(c)] \rightarrow 0$ as $q \rightarrow \infty$: inter-pair standard deviation decreases from 0.54 ($q = 29$) to 0.16 ($q = 101$).*

Proof. Items (i)–(ii) are proved in O22–O23 [3, 6]. Item (iii) is established in O24 [7]. Item (iv) is the main numerical result of O25 [5]: the inter-pair variance of δ_{pair} is reported as $\sigma = 0.54$ at $q = 29$, $\sigma = 0.26$ at $q = 61$, $\sigma = 0.16$ at $q = 101$, decreasing monotonically. $\square$

Remark 3.3 (Scope of [U]). Items (i)–(iii) of Proposition 3.2 establish [U] at the structural level for the sectors $m = \pm 1$ vs. $m = 0$. Item (iv) provides quantitative confirmation that the remaining inter-pair variance vanishes. The full proof of (8) for general c — covering sectors beyond the canonical BI-parity pairing — is the content of the O25 large-prime extension ($q = 211$ confirmed: $A_H(211) = 2.022$, Stage C PASS; $q \in \{307, 401\}$ pending).

4 Asymptotic $\mathfrak{su}(2)$ -Isotropy

4.1 Character-independence implies $\mathfrak{su}(2)$ -invariance

Lemma 4.1 (Asymptotic $\mathfrak{su}(2)$ -isotropy). *Under the O-series universality [U] and the weight–character correspondence (7), the effective quadratic form*

$$Q_q := A_H(q)(k_X^2 + k_Y^2) + A_Z k_Z^2 \quad (9)$$

is asymptotically $\mathfrak{su}(2)$ -invariant as $q \rightarrow \infty$:

$$|A_H(q) - A_Z| \leq C' \varepsilon(q), \quad (10)$$

where $\varepsilon(q) \rightarrow 0$ is the universality rate of [U] and $C' > 0$ is a constant independent of q .

Proof. We relate the anisotropy $|A_H - A_Z|$ to the character variance $\varepsilon(q)$ through three steps.

Step 1: Anisotropy measures character variance. The coefficient $A_H(q)$ is determined by the effective form restricted to the $m = \pm 1$ sectors (horizontal), and A_Z by the $m = 0$ sector (central). This identification follows from Q7 Criterion 5.1 [9], which relates the symbol coefficients of $\sigma_2(L_{\text{eff}})$ to the fingerprint energy sector-wise: the coefficient of $k_X^2 + k_Y^2$ equals the admissibility weight in the $m = \pm 1$ spin sector, and the coefficient of k_Z^2 equals the weight in the $m = 0$ sector. We do not re-derive this identification here, but note that it is the content of Q7 §5.1 and depends only on the bridge non-obstruction established in Q7–Q9 (not on the full bridge conjecture).

Under the weight–character correspondence (7), the ratio A_H/A_Z therefore equals the ratio of fingerprint energy averaged over $m = \pm 1$ characters to the energy at $m = 0$. Note that since the symbol coefficients depend linearly on the admissibility weights (Q7 Criterion 5.1 [9]), sector-wise averages of σ_c directly determine A_H and A_Z , and the ratio is well-defined and finite for all q :

$$\frac{A_H(q)}{A_Z} = \frac{E_c[\sigma_c(n) : c \in m = \pm 1 \text{ sector}]}{E_c[\sigma_c(n) : c \in [q/2 - w, q/2 + w]]}. \quad (11)$$

Step 2: Universality bounds the ratio. Under [U]: $|\sigma_c(n) - \sigma_*(n)| \leq \varepsilon(q)\sigma_*(n)$ for all c and uniformly over the fitting window $n \leq n_*(q)$ (Definition 3.1). Therefore:

$$\left| \frac{E_c[\sigma_c]}{E_{c'}[\sigma_{c'}]} - 1 \right| \leq 2\varepsilon(q) \quad (12)$$

for any two character averages $E_c, E_{c'}$ (since both are within $\varepsilon(q)$ of σ_*). Applying (12) to the ratio in (11):

$$\left| \frac{A_H(q)}{A_Z} - 1 \right| \leq 2\varepsilon(q). \quad (13)$$

Step 3: Absolute bound. Since $A_Z = 2$ (Q8), multiplying through by $A_Z = 2$ gives:

$$|A_H(q) - 2| \leq 4\varepsilon(q) =: C' \varepsilon(q). \quad (14)$$

As $\varepsilon(q) \rightarrow 0$ by hypothesis [U], we conclude $A_H(q) \rightarrow A_Z = 2$. $\square$

Remark 4.2 (Role of BI parity). Proposition 3.2(i) establishes that the $m = +1$ and $m = -1$ sectors are exactly equal ($\sigma_c = \sigma_{q-c}$) for all finite q . This means the form (9) is already isotropic in the horizontal plane: the coefficients of k_X^2 and k_Y^2 are exactly equal for all q . The remaining anisotropy $A_H(q) \neq A_Z$ is a vertical–horizontal asymmetry (between $m = 0$ and $m = \pm 1$), which is driven by the finite- q corrections to the universal power law $\sigma_*(n)$. As these corrections vanish by [U], the full $\mathfrak{su}(2)$ -isotropy is recovered.

5 Casimir Identification and Proof of Theorem 1.1

5.1 Uniqueness of the $\mathfrak{su}(2)$ -invariant form

Proposition 5.1 (Casimir uniqueness on $\text{Sym}^2(V_\rho)$). *The space $\text{Sym}^2(V_\rho)$ carries the spin-1 irreducible representation of $\mathfrak{su}(2)$. The unique (up to positive scalar) $\mathfrak{su}(2)$ -invariant positive-definite quadratic form on $\text{Sym}^2(V_\rho)$ is the Casimir:*

$$C_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 2 \cdot \text{Id}. \quad (15)$$

In particular, the unique $\mathfrak{su}(2)$ -invariant co-metric with spatial eigenvalue $A_Z = 2$ is $g_{\text{sp}} = 2(k_X^2 + k_Y^2 + k_Z^2) = C_{\mathfrak{su}(2)} \cdot (dk_X^2 + dk_Y^2 + dk_Z^2)$.

Proof. By Schur's lemma, the space of $\mathfrak{su}(2)$ -equivariant endomorphisms of an irreducible representation is one-dimensional (scalar multiples of the identity). Every $\mathfrak{su}(2)$ -invariant quadratic form Q on $\text{Sym}^2(V_\rho)$ is associated to such an endomorphism and is therefore a scalar multiple of the Casimir form. The Casimir $C_{\mathfrak{su}(2)}$ acts on the spin- j representation with eigenvalue $j(j+1)$. For spin $j = 1$ ($\text{Sym}^2(V_\rho)$): $C_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 1 \cdot 2 = 2$. This was established unconditionally in Q8 [4]. $\square$

5.2 Proof of the main theorem

Proof of Theorem 1.1. By Lemma 4.1, under [U]:

$$|A_H(q) - A_Z| \leq C' \varepsilon(q) \rightarrow 0. \quad (16)$$

Since $A_Z = 2$ (Q8, Proposition 5.1):

$$|A_H(q) - 2| \leq C' \varepsilon(q) \rightarrow 0. \quad (17)$$

The rate $\varepsilon(q) = O(q^{-\alpha})$ is bounded from the universality rate [U]; numerically $\alpha \approx 0.44$ (Section 6, Table 1).

For the uniqueness part: once $A_H = A_Z = 2$, the effective spatial form $2(k_X^2 + k_Y^2 + k_Z^2)$ is $\mathfrak{su}(2)$ -invariant and equals the Casimir form by Proposition 5.1. By Q7 Lemma 4.3 [9] (rigidity: the bridge φ is unique up to positive scalar), this Casimir form uniquely identifies the bridge as the isometric $\mathfrak{su}(2)$ -equivariant isomorphism $\varphi : H_{\text{eff}} \xrightarrow{\sim} W_{\text{sp}}$ of Q7 Conjecture 4.7 [9]. $\square$

Remark 5.2 (Epistemic status). Theorem 1.1 is conditional on [U], which is:

- (a) proved structurally for the BI-parity sectors ($m = +1$ vs. $m = -1$, exact equality for all q);
- (b) established structurally for the three-direction count ($\Sigma_c(n_3) = 3$, O23);
- (c) confirmed numerically for $q \in \{29, 61, 101, 151, 211\}$ with strictly decreasing inter-pair standard deviation $\sigma_q \in \{0.539, 0.259, 0.161, 0.139, 0.109\}$ (O25 [5]);
- (d) open in full generality (complete analytical proof of (8) for the rate $\varepsilon(q) \rightarrow 0$).

The convergence $A_H(q) \rightarrow 2$ is therefore established at the same level of confidence as the O-series spectral universality: structurally supported and numerically confirmed, with the full analytical proof identified as the remaining open step.

6 Numerical Confirmation

Table 1 records the isotropy gap $|A_H(q) - A_Z|$ from the Q7 numerical campaign [9], together with the O25 inter-pair variance $\text{Var}_c[\delta_{\text{pair}}(c)]$ from the O25 campaign [5].

Table 1: Isotropy gap and O25 inter-pair variance by prime q (sector $c = 7$).

q	$A_H(q)$	$ A_H - 2 $	$\text{Var}_c[\delta_{\text{pair}}]^{1/2}$
61	2.041	0.041	0.259
101	2.019	0.019	0.161
151	2.009	0.009	0.139
211	2.022	0.022 [†]	0.109

[†] At $q = 211$, Stage B (metaplectic overlap) does not pass ($c = 7$, overlap = 0.38, cf. Q7 Remark 6.4); the Stage A eigenvalue ratio is $[1 : 0.74 : 0.74]$ rather than $[1 : \frac{1}{2} : \frac{1}{2}]$, consistent with a trajectory that does not fully explore H_{eff} at $\text{bfs_frac} = 0.11$. Stage C (primary criterion, analytically grounded via Q7 Corollary 6.2) passes with zero cross terms and zero isotropy gap, giving $A_H(211) = 2.022$. Stage B failures of this type are finite-coverage artefacts (cf. Q7 Remark 6.4 for the analogous anomaly at $q = 61$, $c = 3$).

Both quantities decrease monotonically with q , consistent with Lemma 4.1: the universality rate $\varepsilon(q)$ controls the isotropy gap. The empirical fit $|A_H(q) - 2| \approx 0.264 q^{-0.44}$ (Q7 [9]) gives $\varepsilon(q) \sim q^{-0.44}$, consistent with an $O(q^{-1/2})$ upper bound from the BFS convergence rate (Q5b Theorem 3.2 [1]). The O25 campaign has since extended the dataset to $q = 211$ ($\text{Var}_c[\delta_{\text{pair}}]^{1/2} = 0.109$, 105 pairs, $M = 50$ [5]), consistent with the monotone decrease. Runs for $q \in \{307, 401\}$ remain in progress and will allow discrimination between the exponent $\alpha = 0.44$ and $\alpha = 1/2$.

7 Consequences

7.1 Completion of the effective metric

Corollary 7.1 (Isotropic Lorentzian metric). *Under [U] and the Q5a hypotheses, the effective co-metric on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ is*

$$g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2), \quad (18)$$

of signature $(-, +, +, +)$ and Lorentzian, with all three spatial eigenvalues equal to the $\mathfrak{su}(2)$ -Casimir eigenvalue on $\text{Sym}^2(V_\rho)$.

Proof. This is Theorem 1.1 combined with Q5b Theorem 6.1 [1] (Lorentzian signature) and Q8 [4] ($A_Z = 2$). $\square$

7.2 Bridge conjecture: closure

Corollary 7.1 completes the closure of Q7 Conjecture 4.7 [9]. Under [U], the effective symbol satisfies Q7 Criterion 5.1 with $A_H = A_Z = 2$: the equivariant bridge $\varphi : H_{\text{eff}} \xrightarrow{\sim} W_{\text{sp}}$ exists and is isometric with respect to the Casimir inner product. The bridge conjecture is therefore resolved in the isotropic case (Q7 Remark 6.8, Case 1), conditionally on [U].

7.3 The physical interpretation

The identification $A_H = A_Z = 2 = C_{\mathfrak{su}(2)}$ carries a structural message: *the effective spatial geometry of the Cosmochrony framework is not an artefact of a specific choice of normalisation, but is forced by the $\mathfrak{su}(2)$ symmetry of the admissible fibre and the representation-theoretic identity $C_{\mathfrak{su}(2)}|_{\text{spin-1}} = 2$.* Isotropy is not postulated; it is selected by the dynamics.

7.4 Open direction: full proof of [U]

The remaining analytical task is to prove [U] in full generality, i.e., to establish the rate $\varepsilon(q) \rightarrow 0$ from the structure of the O-series rather than from numerical evidence. The natural approach, following the O25 programme, is to:

1. Bound the inter-pair variance $\text{Var}_c[\delta_{\text{pair}}(c)]$ by the convergence rate of the BFS geometry to the Carnot–Carathéodory geometry of $\text{Heis}_3(\mathbb{R})$ (Q5b Theorem 3.2 [1]).
2. Show that this convergence rate is $O(q^{-1/2})$ from the Bass–Guivarc’h polynomial ball growth (Q5b Theorem 2.1 [1]).
3. Conclude $\varepsilon(q) = O(q^{-1/2})$, giving $|A_H - 2| = O(q^{-1/2})$.

This programme would also determine the precise exponent α , resolving the discrepancy between the empirical $\alpha \approx 0.44$ and the structural prediction $\alpha = 1/2$.

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